

James Wanjiku

AI/ML Engineer · LLM systems, voice AI, Python backend

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PROFESSIONAL SUMMARY

AI/ML engineer with two years building and running LLM systems in production. I lead engineering on an autonomous B2B voice-calling platform that handles thousands of concurrent live calls across several languages, built on the OpenAI and Deepgram APIs, a LangGraph conversation engine, FastAPI and GCP. My work runs from prompt and orchestration design through evaluation to deployment and production monitoring. Lead author on a paper about LLM instruction-following, accepted at ICIST 2026 in Kaunas.

SKILLS

Languages: Python, TypeScript, SQL, C++

GenAI / LLM: LLM API integration (OpenAI, Anthropic), prompt engineering, agentic workflows and orchestration, LangGraph, multi-agent systems, RAG, embeddings, hybrid retrieval (BM25 + vector), vector search, LLM evaluation (LLM-as-judge, DeepEval), MCP (Model Context Protocol), latency / cost / retry / fallback handling, speech-to-text and text-to-speech (Deepgram)

Backend: FastAPI, REST APIs, async Python, WebSockets, Pydantic schemas, Redis, PostgreSQL, SQLAlchemy

Infrastructure / DevOps: Docker, CI/CD, GitHub Actions, Google Cloud Platform (GCP), observability and structured logging, pytest, pre-commit, uv, ruff

Computer vision: PyTorch, YOLO, DINOv2, ORB-SLAM3, multi-object tracking, geospatial ray-casting

Other: team leadership, technical and grant-proposal writing, English (full professional proficiency)

EXPERIENCE

AI/ML Engineer and Team Lead, AAI Labs (Vilnius, Lithuania; remote)

Jul 2024 - Present

Team Lead since Jul 2026

- Lead engineering on an autonomous outbound B2B voice-calling platform (OpenAI LLM, Deepgram STT/TTS, FastAPI, GCP, Redis). Took it from single calls to thousands of concurrent live calls across multiple languages, and validated the concurrency ceiling end to end against the telephony provider.
- Built the conversation engine as a LangGraph state machine over the call stages (gatekeeping, discovery, pitch, objection handling, close), with domain guardrails and backtracking, so multi-turn calls stay on script.
- Added automatic language detection so the agent answers in the caller's language; shipped an assisted mode that lets a human operator join or take over a live call.
- Own CI/CD, containerized deployment, and production logging and monitoring for the AI services.
- Led a three-person research team on instruction-following in multi-turn LLM workflows, comparing prompting against explicit control architectures with a parallel background validator, and built the LLM-as-judge evaluation harness (DeepEval). Paper accepted at ICIST 2026, Kaunas.
- Built the georeferencing and object-tracking pipeline for a 5G-connected drone system: YOLO detection and multi-object tracking on live video, each detection ray-cast to GPS coordinates from the drone's position and gimbal orientation. Presented to and accepted by an EU committee.
- Lead the technical concept and writing for AAI Labs' Horizon Europe research grant proposals and coordinate consortium partners across Europe.
- Earlier delivery: NLP duplicate-detection over public-sector funding applications using vector search; an LLM-backed recommendation service on FastAPI and GCP.

Network Administrator, JayBee Internet

Aug 2023 - May 2024

- Maintained network infrastructure and user support before moving into AI engineering.

SELECTED PROJECTS

Personal initiatives, built outside AAI Labs.

Taura 2.0, open-source Kikuyu-English machine translation 2025 - present

- Bidirectional MT engine for a low-resource language with almost no existing tooling. Cross-lingual FastText embeddings aligned with Orthogonal Procrustes, refined with mutual nearest neighbour and CSLS.
- Hybrid retrieval at inference: BM25 keyword search combined with embedding retrieval, which lifts recall on proper nouns and domain terms. FastAPI service over 19,300+ curated sentence pairs.

Nemo Voice Typing, offline on-device dictation, Windows 2026 - present

- Streaming speech recognition running locally on CPU with NVIDIA NeMo, no cloud round-trips. C# and .NET.

PUBLICATIONS

J. Wanjiku, A. Zhurba, Yeabtsega. **Towards Reliable Instruction-Following in LLM Multi-Turn Workflows: An Empirical Comparison of Prompting and Control Architectures.** Accepted, ICIST 2026, Kaunas, Lithuania. Lead author.

EDUCATION

BSc Computer Science, University of Embu, Kenya

2018 - 2023

CERTIFICATIONS

DeepLearning.AI / Coursera: Machine Learning Specialization, Natural Language Processing Specialization, Mathematics for Machine Learning, Introduction to Large Language Models